



It's Not My Responsibility: Working with Autonomy-Restricting Algorithms Facilitates Unethical Behavior and Displacement of Responsibility

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Abstract

While algorithmic decision-making technology can greatly improve efficiency at work, it has the potential to promote unethical decision making in humans who work with the technology. In this paper, we test how the moral disengagement mechanism, displacement of responsibility arises, as an outcome of working with autonomy-restrictive algorithmic decision-making (ADM) technology. We draw upon the tenets of the job characteristics model and moral disengagement theory to test how job autonomy-restrictive ADM affects moral disengagement through displacing an individual's responsibility for unethical behavior. We experimentally manipulated the level of job autonomy afforded by an algorithmic ADM system across two realistic work simulations. In Study 1 ($N=276$), we first showed that participants working with autonomy-restricting ADM engaged in higher rates of unethical behavior that disadvantages customers to the benefit of the organization, compared to those who work with autonomy-affording ADM. In Study 2 ($N=256$), we used a novel measure of the cognitions associated with displacement of responsibility to show that working with an autonomy-restricting ADM system leads individuals to experience subjectively lower levels of personal responsibility for the negative consequences of their decisions and leads them to displace this responsibility onto a figure of authority, which can be a human or the technology itself. Altogether, our findings provide new insights into how the cognitive process of moral disengagement unfolds, and indicate that job autonomy, as afforded or restricted in an algorithmic decision-making context, is an important consideration when implementing ADM technology.

Keywords Algorithms · Moral disengagement · Work design · Unethical behavior · Responsibility

Introduction

We have witnessed the accelerated use of digital technologies in work. One of the major attractions of introducing such technologies into organizations is that work tasks can be completed with improved efficiency (Bader & Kaiser, 2019; Gombolay et al., 2015; Huang & Rust, 2018). Algorithmically aided decision making, which refers to “automated decision making based on predefined rules and/or goals” (Lindenbaum et al., 2020, p. 247) is increasingly

being adopted in sectors such as health, customer service, and finance (Huang & Rust, 2018). In organizational settings, algorithmic decision making (ADM) involves using algorithms to support, inform, or govern decisions pertaining to organizational functioning (Leicht-Deobald et al., 2019). A burgeoning area of research focuses on how individuals think about the moral consequences of working with ADM (Ebrahimi & Matt, 2023; Nichol et al., 2024; Shank et al., 2019). There is a growing focus on understanding how employees might morally disengage from unethical consequences arising from working with ADM technology (Ebrahimi & Matt, 2023; Nichol et al., 2024).

Moral disengagement refers to the deactivation of self-regulatory processes that typically keep unethical behavior ‘in check,’ and in doing so, individuals can rationalize their unethical behavior as acceptable (Bandura, 1986). Meta-analytic research has shown moral disengagement to be associated with unethical behavior in organizations

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(Newman et al., 2020; Ogunfowora et al., 2021). Yet, further research is needed to better understand how moral disengagement unfolds under particular organizational contexts, including those in which employees are asked to work with ADM. Despite much-lauded advantages of ADM, there is now agreement among scholars that its implementation has the potential to result in negative consequences (Gratch & Fast, 2022; Sullivan & Fosso Wamba, 2022; Wesche & Sonderegger, 2019). Recent research shows that humans tend to endorse unethical ADM recommendations and they can morally disengage from these decisions by using the moral disengagement mechanism, displacement of responsibility (Ebrahimi & Matt, 2023), which refers to shifting moral responsibility for unethical decisions from the self to a more plausible figure of authority (Bandura, 1986). This phenomenon is important to understand because it can explain why individuals continue with unethical behavior over time. In a nutshell, moral disengagement disrupts processes that might otherwise have led an individual to behave *ethically* by self-regulating their actions, and it is important to understand how a potential breakdown in this process of self-regulation occurs.

Despite the importance of this phenomenon, previous research empirically examining moral disengagement (within or beyond the context of working with ADM) has not actually unpacked how the process of responsibility actually *shifts* from the self to another entity, such as a human supervisor or the technology itself. Our goal in this paper is to examine this process. We first investigate how the autonomy of employees who use ADM can affect the incidence of unethical behavior, defined as engaging in behavior that is contrary to accepted moral norms in society (Treviño et al., 2014). Unethical behavior in the context of ADM can include disadvantaging particular groups of people like an organisation's clients/customers, to meet the goals of organizational efficiency (Marabelli et al., 2021). These types of acts are examples of unethical pro-organizational behavior, which is defined as "actions that are intended to promote the effective functioning of the organization or its members (e.g., leaders) and violate core societal values, mores, laws, or standards of proper conduct" (Umphress & Bingham, 2011, p. 622). To the best of our knowledge, only one (cross-sectional) study examined the role of job autonomy in facilitating unethical pro-organizational behavior (Kalshoven et al., 2016). Here, low levels of autonomy were positively associated with employees' willingness to engage in unethical pro-organizational behavior. However, this study did not examine a causal relationship, nor did it investigate moral disengagement as a variable of interest. Furthermore, prior research examining autonomy and unethical behavior has focused on counterproductive work behavior or employee deviance (e.g., Fida et al., 2015; Fox et al., 2001; Lu et al., 2017), which centers on employees engaging in behavior

that is detrimental (as opposed to helpful) to their organization. We second investigate how—having behaved unethically—moral disengagement might then occur. To allow strong internal validity, we investigate these questions in two simulation studies in which we vary the degree to which an ADM system offers autonomy to participants, and then investigate whether variations in this autonomy shape participants' unethical behavior, as well as how people react after being unethical.

We draw upon work design theory and moral disengagement theories to conduct our investigation. Work design refers to the content and organization of one's work tasks, relationships, roles, and responsibilities (Parker, 2014). Over 100 years of research has shown work design to play a crucial role in fostering many important work-related outcomes such as wellbeing, performance (Parker et al., 2017), learning, and cognition (Parker et al., 2021). Specifically, we focus on an individual's level of job autonomy, or the extent to which one has control over their work and can decide when and how to perform it (Hackman & Oldham, 1976). Job autonomy is a central concept in the classic Job Characteristics Model (Hackman & Oldham, 1976) that is concerned with creating psychologically motivating work roles. Autonomy has several positive psychological consequences, such as fostering a sense of ownership and allowing people to more actively manage work demands (Parker, 2014).

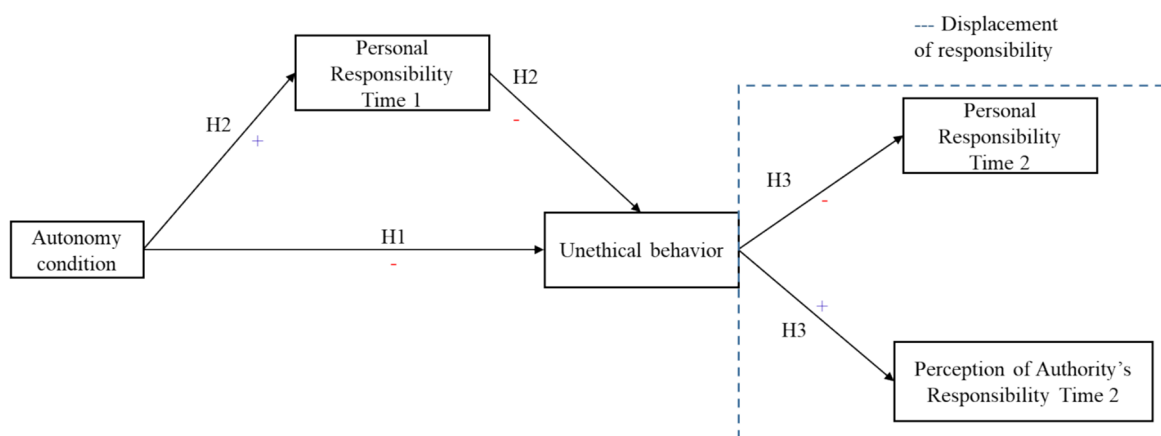
Altogether, therefore, we propose that, in the context of working with ADM, a critical consequence of reducing job autonomy is that individuals feel less personally responsible for the work they do, and its outcomes (Hackman & Oldham, 1976; Humphrey et al., 2007). Under conditions of low autonomy, employees are not in a strong position to choose the methods by which they complete their work, when they complete specific tasks at work, or engage in significant decision making. Individuals will experience lower personal responsibility for their actions, which can in turn predict unethical behavior. From a technological perspective, our prediction aligns with scholars who have advocated for workers to be able to control and influence technology. For example, in their analysis of algorithmic management impact on human workers, Parent-Rocheleau and Parker (2022) recommended that some of the negative motivational impacts of algorithmic management on workers might be reduced if workers had some degree of control and influence over the system. A laboratory study by Dietvorst et al. (2018) supports this reasoning, showing that even a small amount of empowerment can overcome the aversive effects of some algorithmic systems.

Crucially, as well as examining the role of job autonomy in predicting unethical behavior, we also draw on moral disengagement theory to theorize what happens *after* an individual has engaged in unethical behavior. Specifically, we argue that individuals who experience low personal

Second, although studies to date have shown how specific work contexts lead employees to use moral disengagement (Barsky, 2011; Huang et al., 2019; Kish-Gephart et al., 2014; Moore et al., 2019), researchers have rarely systematically examined moral disengagement through the

Finally, we contribute more specifically to understanding the process of displacement of responsibility—much discussed, but rarely operationalized (Schaefer & Bouwmeester, 2021). We provide a novel approach to measuring moral disengagement by using a unique experimental paradigm and a theoretically-driven approach to examine the cognitions associated with the displacement of responsibility. In so doing, our research answers several calls for a deeper understanding of process moral disengagement, which requires investigations that allow researchers to isolate the components of each specific moral disengagement mechanism (Moore, 2015; Newman et al., 2020; Schaefer & Bouwmeester, 2021).

Algorithms are frequently used to improve the efficiency of decision making. By processing relevant data available to the algorithm and using those data to quickly present employees

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with choices on how to respond to a situation, algorithmic recommending reduces or eliminates the need for employees to think through decisions and their implications, thereby fostering efficiency (Kellogg et al., 2020, p. 372). However, not all ADM-based recommendations are necessarily ethical (Martin, 2019)—they can provide recommendations that discriminate or unfairly disadvantage individuals (Ebrahimi & Hassanein, 2021; Ebrahimi & Matt, 2023). Well-known examples include the ‘Robodebt’ scandal in Australia which involved hundreds of thousands of welfare recipients being deemed to owe a debt to the government that was calculated by an algorithmic decision-making system with little to no human oversight (Holmes, 2023; Lewis, 2023). The Robodebt system has been the subject of the highest form of public inquiry in Australia, a Royal Commission. Another example is that of rideshare companies, such as Uber or Lyft, which use algorithmic management systems to perform multiple functions, from task allocation to contract termination. In recent years, the algorithmically managed food delivery company, Deliveroo, was successfully sued for unfair dismissal of an employee based on the algorithm’s decision that the driver in question delivered food too slowly (Veen et al., 2021).

As multiple authors have discussed, technology does not have a deterministic impact on outcomes (e.g., Parent-Rocheleau & Parker, 2022; Strohmeier, 2009). Rather, the impact depends on how that technology is designed and implemented, and the choices made by various stakeholders. If the way in which the technology is implemented into the organization affords employees the autonomy to decide whether they respond to a situation using the suggestion provided by the software, they would be working with what we call autonomy-affording ADM. These employees are able to choose whether to comply entirely with the recommended suggestion, take on board aspects of the suggestion, or to disregard the suggestion entirely, especially if it does not represent the best option for their clients. Under such a scenario, we would expect that employees will be more likely to avoid actioning recommendations that might harm clients. In contrast, we expect that employees who are not actively encouraged to exercise their autonomy, that is work with autonomy-restrictive ADM, may opt for the easier and more efficient option of accepting the recommendation decision from the algorithm, irrespective of its consequences for clients. We therefore propose:

Hypothesis 1 *Autonomy-restrictive ADM is positively associated with unethical behavior, whereas autonomy-affording ADM is negatively associated with unethical behavior.*

To explain why autonomy-affording ADM is likely to lead to more *ethical* behavior compared to autonomy-restrictive ADM, we turn to the Job Characteristics Model

(JCM; Hackman & Oldham, 1976). The JCM posits that employees who experience sufficient levels of positive work characteristics will subsequently experience critical psychological states that lead to desirable outcomes for the employee and organization (e.g., job satisfaction and performance). Within the JCM, job autonomy fosters the psychological state of personal responsibility for the outcomes or consequences of one’s work (here onwards we refer to this state as ‘personal responsibility’). In other words, individuals will recognize that outcomes of the job “depend increasingly on the individual’s own efforts, initiatives, and decisions rather than on the adequacy of instructions from the boss or on a manual of job procedures,” leading to “strong personal responsibility for the success and failures that occur on the job” (Hackman & Oldham, 1976, p. 258). This greater personal responsibility and sense of ownership of outcomes means that individuals are likely to think more widely about any repercussions of their action, and, in line with moral disengagement theory, be more likely to activate their self-sanctions against unethical behavior (i.e., choose a more ethical course of action) because they perceive they are responsible for the consequences of their own actions (Bandura, 1986). On the other side of the coin, employees who experience low levels of job autonomy will experience low levels of personal responsibility. As such, they will be less likely to take ownership or responsibility for any unethical behavior because they do not see themselves as the agent responsible for suggesting an unethical course of action. This argument is consistent with Parker (2014) who speculated that low job autonomy might deter employees from taking moral responsibility at work because of their narrow “not my job” role orientation means they do not feel ownership of the situation. We therefore hypothesize:

Hypothesis 2 *The positive association between autonomy-restrictive ADM and unethical behavior is mediated by a low level of personal responsibility.*

The JCM’s focus on personal responsibility provides a theoretical basis for understanding whether and why an unethical behavior occurs in the first place. We go further in this research and apply moral disengagement theory to understand what happens when a decision to behave unethically has been enacted (Bandura, 1986). That is, whereas Hypothesis 2 focused on low personal responsibility as an antecedent of unethical behavior, here we focus on how people explain their actions to themselves after they make an unethical decision. This focus is important because, as we elaborate below, empirical research that explicitly tests the core theoretical assumptions of how moral disengagement unfolds when individuals actually engage in unethical behavior is very scarce.

According to moral disengagement theory, people are intrinsically motivated to maintain a positive moral self-image that adheres to personal and social standards of moral behavior, which is achieved through internal self-regulation. Behaving in ways contrary to one's personal or social standards (i.e., unethically) leads to unpleasant emotional or cognitive consequences. Yet, people are often faced with opportunities to behave unethically and, when the motivation to do so is strong enough, Bandura suggested that people can deactivate their self-regulatory processes using moral disengagement. By morally disengaging, individuals rationalize their behavior as acceptable in relation to their internal standards or social context using specific cognitive mechanisms. Moral disengagement is said to occur “most prominently in patterns of behavior that serve the user in some way but injure others” (Bandura, 1986, p. 375). Bandura described eight moral disengagement strategies, however, only one strategy need be available to the individual. In this research, for reasons we outline below, we focus on the specific strategy of displacing responsibility.

The role of displacement of responsibility in relation to employee unethical behavior when working with ADM technology has been of increasing interest to researchers in recent years. This is unsurprising given the scale at which algorithms are being deployed to oversee or influence the work of human employees, which by design influences workers' sense of control (Bankins & Formosa, 2023; Parent-Rochelleau & Parker, 2022; Parker & Grote, 2022). In general, studies show displacement of responsibility to play an influential role in influencing unethical behavior in the context of ADM. There is some emerging research on how software developers perceive the ethical implications of the algorithmic technology they develop. In a qualitative sample of healthcare AI developers, Nichol et al. (2024) found that while some developers do understand that their technology has the potential to cause harm, there can be uncertainty about who is responsible for preventing harm when working in development companies that prioritize profits. In another study, Gao et al. (2024) found in a time-lagged survey that gig workers who reported higher perceptions of algorithmic monitoring were more likely to engage in unethical behaviors like careless deliveries, and this was mediated by a trait-like tendency to displace responsibility at work. When examining how workers make decisions when working with ADM technology in an experimental study, Ebrahimi and Matt (2023) found that employees faced with making complex decisions can use displacement of responsibility as a key moral disengagement mechanism when intending to approve the unethical recommendation of data analytics algorithms. In both these quantitative studies, humans experienced low levels of control over their work, which activated a contextually-relevant tendency to displace responsibility, however, the explicit role of personal responsibility in relation to that

of an authority figure was not considered, which we suggest is essential to do, given how displacement of responsibility is conceptualized.

According to Bandura's conceptualization, there are three necessary ingredients for displacing responsibility to have occurred. First, the individual must either perceive some imperative and opportunity to commit an unethical act or have already committed an unethical act. If there is no unethical act, there is no need to displace responsibility. Second, the individual should experience low levels of personal responsibility for the consequences of their actions, such that they do not feel they are the true agent or controller of their actions. Third, the individual should direct the responsibility for the unethical behavior or its adverse outcomes onto some figure of authority that creates an imperative (implicitly or explicitly conveyed) for that individual to engage in unethical behavior. Displacement of responsibility is said to occur when individuals who do not feel in control of their behavior perceive that a figure of authority is responsible for their behavior and that behavior's effects on others. Thus, to the extent that technologies such as ADM ‘blur’ the entity that should be ultimately responsible for negative outcomes of a decision made (Kellogg et al., 2020; Martin, 2019), displacement of responsibility for consequences of that decision becomes more cognitively plausible. As we explain next, in the present research, we propose that employees who work under autonomy-restricting ADM are more likely to subsequently reconcile their actual unethical behavior if they feel that a figure of authority is responsible for it, and consequently may be more likely to engage in it more frequently.

There are at least three candidate figures of authority in the context of working with ADM that could plausibly be perceived by employees as having more control over their work than themselves. First, a human higher-level manager, with the unethical agent focusing on how manager/executives had decided how to implement the ADM into an organization (Kellogg et al., 2020). Second, the ADM technology itself might be perceived as a proverbial figure of authority, given that the role of this technology is to provide employees with ostensibly legitimate courses of action to take in response to a situation. Third, the ADM designers (or developers) might be considered responsible because they had programmed the ADM suggestions, without adequately accounting for any consequences (Mittelstadt et al., 2016; Sullivan & Fosso Wamba, 2022). Thus, if an employee working with autonomy-restricting ADM feels they should enact an ADM recommendation that constitutes unethical behavior, that employee might displace responsibility for the unethical behavior and its consequences onto a manager, the ADM technology per se, and/or its designers.

Capturing the process of moral disengagement in situ has proven difficult for researchers (Moore, 2015). To date, researchers have used various methods including

contextualized measures of the trait-like tendency to morally disengage (e.g., Huang et al., 2018), asking participants post hoc why they made unethical choices in the contexts of experimental tasks (Reynolds et al., 2014), or observing individuals engaging in unethical behaviors that contradict their prosocial values (Shepherd et al., 2013). However, none of these approaches specifically speak to the individual cognitive components of displacement of responsibility. There is some research that tries to capture contextually activated displacement of responsibility (Antunez et al., 2024; Barsky, 2011; Hinrichs et al., 2012; Thiel et al., 2023) but these studies use adapted propensity measures of moral disengagement and do not unpack the shift in responsibility from oneself to figures of authority. Thus, here, we develop a novel work-simulation paradigm that is designed to activate and capture all three components of displacement of responsibility—and to experimentally test the causal relation between autonomy and displacement of responsibility. We hypothesize that:

Hypothesis 3 *When an individual has engaged in unethical behavior under conditions of low autonomy, they are less likely to experience subsequent personal responsibility for the outcomes of one's work (3a), and more likely to subsequently attribute responsibility onto a figure of authority (i.e., displace responsibility) for the outcomes of one's work (3b).*

Importantly, we adopt an experimental approach to testing our hypotheses as it allowed sufficient internal validity to isolate the causal role of work design. Participants completed a realistic email inbox task (Barsky, 2011; Treviño & Youngblood, 1990) as customer service officers in a fictitious hotel either under autonomy-affording or autonomy-restricting ADM. The context of the service industry also adds realism to the task as ADM has been used in the service industry for a number of years now (Huang & Rust, 2018). Participants engaged in the task under randomly assigned conditions with either autonomy-affording or autonomy-restricting ADM. Our experimental approach circumvents many of the challenges of investigating our focal constructs of interest in a field context, including difficulty in controlling for extraneous contextual factors that could influence participants' perception of job autonomy or ambiguity over cause and effect relationships over time (Podsakoff & Podsakoff, 2019). We further answer multiple calls to undertake experimental research when studying moral disengagement (Newman et al., 2020; Schaefer & Bouwmeester, 2021), not just to understand causal relationships, but to reduce the social desirability that is often associated with self-report field study measures of unethical decision making. Study 1 aims to test the first two hypotheses. In Study 2, we made

some improvements to the core experimental paradigm and our measures in order to test all three hypotheses.

Study 1

Method

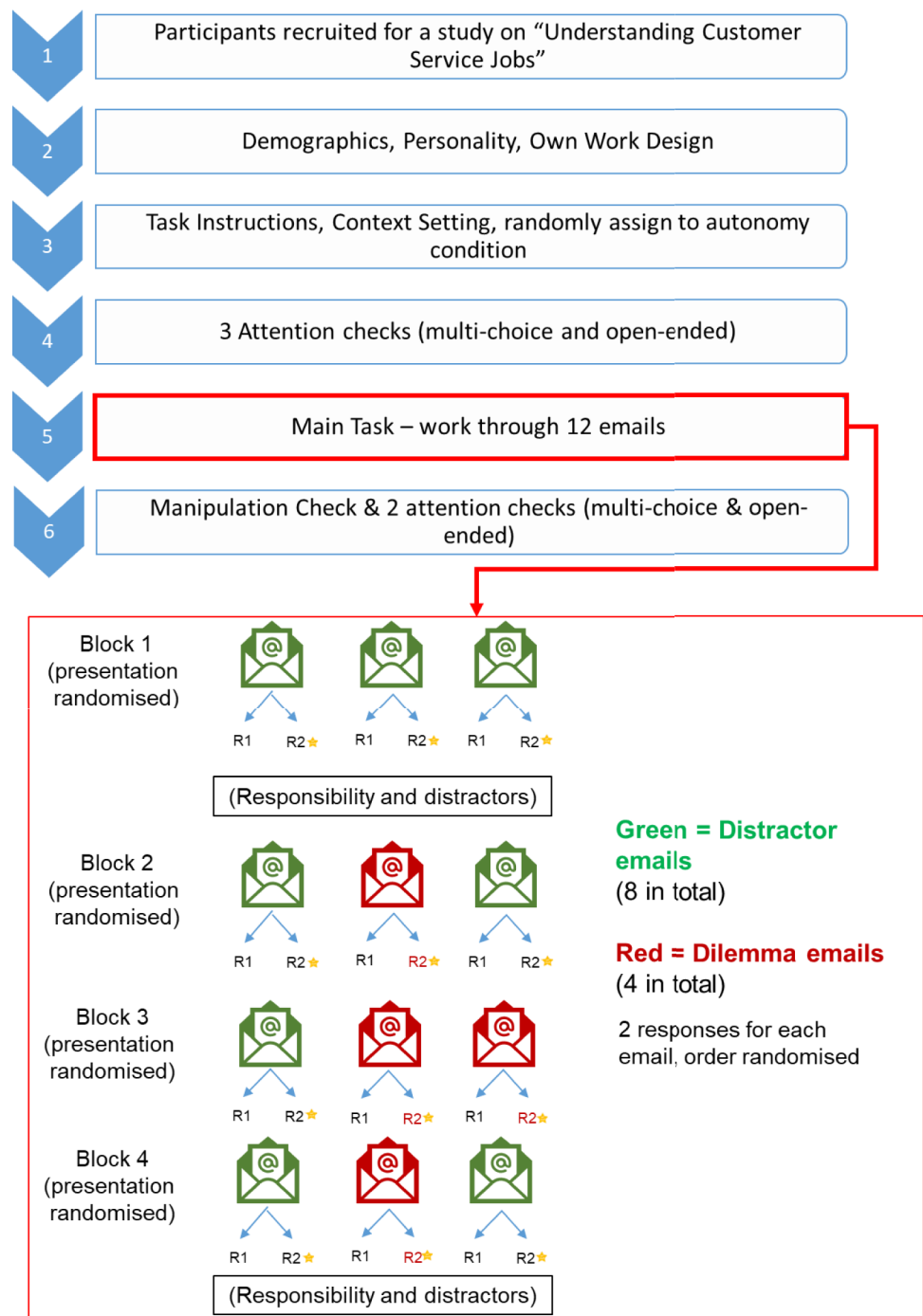
All studies received approval from the university's institutional Human Research Ethics Committee (approval number: RA/4/20/4433). To mask the purpose of this research, we advertised all studies to participants under the title "Understanding Customer Service Jobs." The online supplement contains three 'mini-studies' for the validation of the novel manipulations and stimuli that were used in all our experimental studies. All recruitment/task instructions, materials, data, and analysis syntax, for the studies conducted in this paper are available on the Open Science Framework (link: https://osf.io/phg82/?view_only=4f847ddbc31d490b90917809c4e07c38). This study was pre-registered on the Open Science Framework. The preregistration for Study 1 is available here: https://osf.io/kw9bg/?view_only=0f11832d566d4966839bc95e0315e0e4 and the online supplement details where and why deviations occurred.

The sample size for both studies was estimated via a priori power analyses (see online supplement) using GPower 3.1 (Faul et al., 2007), which indicated a required sample of 210 participants. However, we intended to operationalize our dependent variable as a count variable for hypothesis testing and Cox and MacKinnon (2010) suggest that mediation analyses using a count-dependent variable may yield biased parameter estimates. They recommend increasing the sample size slightly to attenuate potential biases hence, we aimed recruit at least 50 additional individuals.

A total of 289 individuals (57% female) from Amazon's MTurk platform were recruited via CloudResearch (Litman et al., 2017). All participants were residents of the United States of America. Using our pre-registered data screening criteria (details in online supplement), we excluded 13 participants, leaving a final sample of 276. In the final sample, participants' mean age was 39.6 years (SD = 11.60) and 88% of participants were in either full-time, part-time, or casual work.

We designed and hosted on the university's Qualtrics platform; see Fig. 2 for a flowchart of the procedure. Following informed consent, participants answered a pre-experiment questionnaire before commencing the main study, which measured demographics and personality. Participants were asked to play the role of a Customer Services Officer (CSO) working at the (fictional) Paradise Point Hotel, where they would respond to a series of emails pertaining to customer booking enquiries, requests for information, or billing issues. As part of their role, participants would work using

Fig. 2 Flowchart of the Procedure for Studies 1 and 2. R1 and R2 refer to each response option; R2 in Red font represents an unethical response option. Yellow star represents a “top reply” response option corresponding to each respective email



a fictitious ADM system called “SmartAnswer™.” ADM is being increasingly deployed in the service sector (Huang & Rust, 2018) and responding to emails from past or future guests is common in the hospitality sector in the twenty-first century. Thus, this task allowed us to create a situation where we expected to observe unethical behavior in a highly naturalistic setting, while maintaining experimental control. To create a realistic work context, an organizational chart showed participants’ job role in relation to others in Paradise

Point Hotel. Participants were informed that at various times during the task, we would ask them how they were finding the job of a CSO.

At the beginning of the task, participants reviewed a worked example showing how to respond to customers using the SmartAnswer™ ADM software. The example explained that each customer email would be ‘analyzed’ by the SmartAnswer™ software, and that the software would present participants with two possible response options. One

of the two options would be marked as a “top reply” with a yellow star next to it, to signify the company-preferred ADM-selected answer. Participants needed to choose which of the two options they would like to use to respond to the customer’s query. The chosen response was then “pasted” into a formatted email with the Paradise Point Hotel signature, which participants could examine before “sending” the email to the customer. After the worked example, the survey software randomly assigned participants to either the high- or low-autonomy condition to complete the inbox task. The manipulation took the form of an email from the Guest Services Manager Riley Watson (i.e., the participants’ hypothetical boss); see the Materials section. After reading the email, participants were asked to type a short reply to Riley Watson, and complete two multiple-choice attention check questions.

All participants worked through 12 customer emails (the stimuli used in our study) across four email blocks, with each block comprising three emails from customers. The situations that were selected for the emails were based on typical traveler experiences seen on websites such as reddit or TripAdvisor, where travelers provide reviews or discuss their experiences. Four emails contained an ethical dilemma (‘dilemma’ emails) and the remaining eight emails served as distractors (i.e., had no ethical valence). The emails within each block were presented in a random order by the survey software, but the blocks were presented in a fixed order.

The first email block comprised three distractor emails to familiarize participants with the task. The next three blocks each contained either one or two dilemma emails. The algorithmically recommended “top reply” response option for the dilemma emails was always associated with an unethical response, whereas the alternative to the top reply option was not unethical. An unethical response was one that allowed the participant as the CSO to engage in behavior consistent with unethical pro-organizational behavior; that is, the CSO was able to choose a response that would prioritize revenue for the organization by disadvantaging the customer financially. Our online supplement details the validation of each email stimulus scenario and corresponding ethical/unethical option. This included a separate set of participants rating the extent to which each response option for an email was perceived as the CSO to be meeting their organization-set key performance indicators (KPIs), while appearing to be unethical by disadvantaging the customer. We calculated mean scores of unethicalness (rated on a Likert Scale of 1—not unethical at all; 2—somewhat unethical; 3—very unethical; 4—extremely unethical) and KPI effectiveness (using a three-point categorical scale: negative effect on performance indicators, No effect on performance indicators, Positive effect on performance indicators). For unethical target responses to be retained, we specified a minimum mean unethicalness rating of 2.5, and a mean KPI effectiveness

rating of 2.5. Additionally, the ethical target responses must have received a mean unethical rating of 2 or lower, and the KPI effectiveness rating must be lower than that of the unethical target response for a dilemma stimulus. Altogether, all unethical dilemma responses were deemed¹ to have met the criteria for signaling unethical behavior relative to the alternative response. For the distractor emails, the “top reply” response usually described a response that logically would help to meet key performance indicators (KPIs), while the alternative either did the same, or did not meet KPIs as clearly as the “top reply.” In addition to serving as distractors, the distractor emails also afforded credibility to the “top reply,” in that the suggested response would generally seem to serve the organization’s goals well. The order in which response options for each email were presented was also randomized. Participants in *both* conditions were *free to deviate* from the SmartAnswer™-suggested responses.

After the first four distractor emails—and again after working through all 12 emails—participants were asked to assess their experience in the job of a CSO. We use “T1” and “T2” labels to distinguish between these two measurement points. The assessment contained a measure of personal responsibility along with other ‘distractor’ measures (see below). At the end of the task, participants answered manipulation check questions and a more attention check questions. Participants were then thanked for their time and paid US\$6.

Materials

Control Variables

Participants completed the 60-item HEXACO Personality Inventory (Ashton & Lee, 2009). We measured two personality traits as control variables, honesty-humility and conscientiousness, as they were identified as potential confounders of the mediator personal responsibility–unethical behavior relation. Previous research shows honesty-humility is negatively associated with unethical behavior (Zettler & Hilbig, 2010), while conscientiousness is positively associated with feeling responsible for work (Ashton & Lee, 2007).

Manipulation of Autonomy

In both the autonomy-restricting and autonomy-restricting conditions, Guest Services Manager Riley Watson first informs CSOs of Paradise Point Hotel’s recent investment in an artificial intelligence system for the purposes of lifting

¹ The unethical target response to the dilemma stimulus “Surprise check-out fee” was rated slightly below the cut-off criterion for unethicalness (mean of 2.33), however, we decided to include this response in the final study given that it had a high KPI effectiveness rating, and that the corresponding ethical target response was rated to be low on unethicalness and KPI effectiveness.

Table 1 Specific wording of autonomy condition manipulation for Study 1

Condition	Wording
Autonomy-affording ADM condition	When responding to customer emails, SmartAnswer™ will recommend a “top reply” to use (denoted by: ) but you can always use your personal initiative and judgment in choosing a response. You are free to make your own decisions
Autonomy-restrictive ADM condition	When responding to customer emails it is in your best interests to  the “Top Reply” option (denoted by: ) out of what SmartAnswer™ provides you with. Your management team advises against deviating from the top reply

the rankings of the hotels in customer service, booking success, and revenue. Riley Watson reminds the CSOs that their performance ratings depend on their booking success and bringing in revenue; this statement was included to create some sense of self-interested incentive to behave unethically, while still benefiting the organization. Autonomy affordance was manipulated in the second part of Riley Watson's email by informing participants in each condition on the extent to which they can make their own decisions (see Table 1) about choosing a response to the customer emails. We based the manipulation text on the wording used in Morgeson and Humphrey's (2006) measure of job autonomy. In the autonomy-affording condition, participants were told that they can use their personal initiative and judgment to answer the queries, while in the autonomy-restrictive condition, participants were not instructed to use their personal initiative and judgment, but instead told that it was in their best interests to use the “top reply” option when responding to customer emails. Both emails had a similar word count to ensure cognitive load in terms of reading was similar in both instances.

Personal Responsibility and Distractor Measures

We adapted four items from the Hackman and Oldham's (1980) ‘experienced responsibility’ measure. Participants were asked to rate the items with respect to their role as CSO at Paradise Point Hotel using a five-point Likert scale (1—*Strongly Disagree* to 5—*Strongly Agree*). The items were “I feel I should personally take the credit or blame for any consequences of our decisions in this job,” “I feel responsible for ensuring that revenue is brought in for the hotel,” “I feel a high degree of personal responsibility for the work I do on this job,” and “I feel responsible for ensuring that the customers' needs are met.” The distractors were single items that asked about task enjoyment, self-efficacy, task difficulty, and job meaningfulness.

Attention and Manipulation Checks

Two attention check items were placed in the HEXACO-60 section of the pre-experiment questionnaire (e.g., “This is an attention check. Please select Disagree”). Three more were presented after Riley Watson's email; participants wrote a short response to Riley's email then answered two multiple-choice questions about: (1) their role in the company and (2) how they were told to respond to customer emails (i.e., use their personal judgment, or advised against deviating from the recommended answer).

At the end of the task, participants answered manipulation and a final attention check. For the manipulation check, participants indicated the extent to which they felt they were able to make their own decisions in responding to customers using a five-point Likert scale, ranging from 1 (*Strongly Disagree*) to 5 (*Strongly Agree*). The attention checks asked participants to choose which out of four statements reflected the study's purpose (e.g., “some questions referred to hotel bookings” and “I was provided information about physical activities that help women to stay healthy”). Finally, participants were asked to write a very brief description of the task.

Results

An independent-samples *t*-test on the manipulation check item indicated that participants in the autonomy-restricting condition ($N = 136$, mean = 2.72, $SD = 1.20$), reported significantly lower levels of autonomy compared to those in the autonomy-affording condition ($N = 140$, mean = 4.10, $SD = 0.72$), ($M_{diff} = 1.38$, $t(220) = -11.51$, $p < 0.001$, Cohen's $d = 1.39$).²

We created an ‘unethical behavior’ count variable using the number of times (out of the four opportunities) that participants chose the unethical response option (i.e., the SmartAnswer™ “top reply”) when responding to dilemma emails. Table 2 provides the percentage of participants that chose the “top reply” option in each condition. The table shows that participants in the autonomy-restricting condition were roughly two to four times more likely than those in the autonomy-affording condition to choose the SmartAnswer™ option for the dilemma email stimuli. However, for most of the distractor stimuli, participants in the autonomy-restricting condition were almost equally likely to choose the “top reply” as those in the autonomy-affording condition; the variation between conditions for some distractor stimuli likely reflects the nature of the autonomy-affording manipulation whereby participants could make their own decisions about email responses.

² Levene's test of equality of variances was violated ($F = 84.90$, $p < .001$), hence, we report the *t*-test that accounts for the violation.

Table 2 Percentage of participants choosing the “Top Reply” option in each Study, grouped by autonomy condition

Type of email	Summary of email	% of participants in low-autonomy condition that chose Top Reply		% of participants in high-autonomy condition that chose Top Reply	
		Study 1 N=136	Study 2 N=132	Study 1 N=140	Study 2 N=124
Dilemma 1	Loyalty points not applied	48	39	12	5
Dilemma 2	Not given the right room	44	33	11	7
Dilemma 3	Surprise booking fee	55	53	25	22
Dilemma 4	Surprise check-out fee	70	58	45	22
Distractor 1	Asking for baby crib	88	79	71	43
Distractor 2	Extending stay at hotel	99	97	100	96
Distractor 3	Loyalty program sign up	91	88	62	53
Distractor 4	Pre-check-in luggage storage	93	83	64	47
Distractor 5	Query about breakfast *	95	14	86	6
Distractor 6	Refund not received *	88	9	61	2
Distractor 7	Shuttle Booking Ticket	96	94	94	91
Distractor 8	Website booking error	99	96	97	91

*denotes that a nonsensical “Top reply” was associated with the distractor stimuli in Study 2

Table 3 Descriptive statistics and correlations among Study 1 variables

Variable	Mean	SD	1	2	3	4	5	6
1. Autonomy condition ^a	NA	NA						
2. Unethical behavior count	1.54	1.49	– .41**					
3. Personal responsibility _{T1}	3.88	0.77	0.31**	– 0.22**	(0.75)			
4. Personal responsibility _{T2}	3.87	0.81	0.31**	– 0.36**	0.81**	(0.81)		
5. Conscientiousness	3.93	0.56	0.10	– 0.02	0.25**	0.21**	(0.80)	
6. Honesty-humility	3.60	0.72	0.09	– 0.13*	0.13*	0.12*	0.22*	(0.81)

N=276. * $p < .05$, ** $p < .01$. Cronbach’s coefficient alpha is provided along the diagonal in parentheses.

^aAutonomy-restrictive ADM=0, autonomy-affording ADM=1

Table 3 shows the descriptive statistics and correlations among study variables. Of note, autonomy-restrictive ADM was associated with higher counts of unethical behavior and lower levels of personal responsibility. Further, only, honesty-humility (not conscientiousness) was negatively associated with *both* unethical behavior and personal responsibility; hence, we only controlled for honesty-humility in the hypothesis tests.

To test whether autonomy-restricting ADM was associated with higher levels of unethical behavior (Hypothesis 1) and whether this relationship was mediated by low levels of personal responsibility (Hypothesis 2), we specified the path model shown in Fig. 3, using Mplus version 8.3. The model incorporates the following paths: (a) the regression of unethical behavior on autonomy condition and personal responsibility at T1 and (b) the indirect path of autonomy to unethical behavior, through personal responsibility at T1. The indirect path was estimated by obtaining 95% bias-corrected confidence intervals derived from 10,000 bootstrapped samples (Hayes, 2018). We also

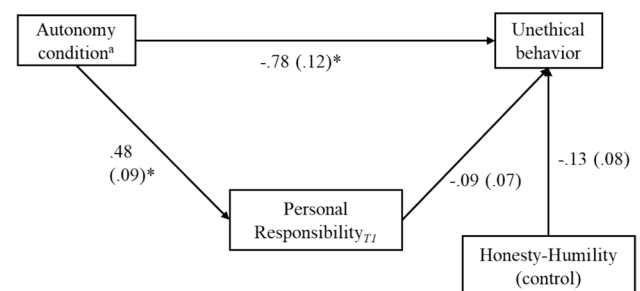


Fig. 3 Path model evaluating the indirect effect of autonomy condition on unethical behavior through personal responsibility_{T1}. Path coefficients are unstandardized, accompanied by standard errors in parentheses. *denotes $p < .001$; all other paths are non-significant. ^aAutonomy-restrictive ADM=0, autonomy-affording ADM=1

controlled for the effects of honesty-humility on unethical behavior.

Before testing our model, we tested the assumption of equidispersion to investigate the appropriateness of

estimating the unethical behavior (a count-dependent variable) pathways using Poisson regression (Coxe et al., 2009). The assumption was violated, as the overall mean ($\mu = 1.54$) and variance ($\sigma^2 = 2.24$) of the unethical behavior count variable in the final sample indicated overdispersion of the data. Accordingly, we specified the unethical behavior regression paths to be estimated using a negative binomial distribution, a recommended alternative to test a model with overdispersed count-dependent variables (Hilbe, 2011).

Overall, Hypothesis 1 was supported, but Hypothesis 2 was not. Results showed a direct negative relationship from autonomy condition to unethical behavior ($B = -0.78$, $SE = 0.12$, $p < 0.001$) and that autonomy condition was positively associated with personal responsibility ($B = 0.48$, $SE = 0.09$, $p < 0.001$) yet personal responsibility was not, in turn, associated with unethical behavior ($B = -0.09$, $SE = 0.071$, $p = 0.195$). The confidence intervals for the indirect pathway from autonomy to unethical behavior crossed zero (indirect effect estimate = -0.044 , 95% CI = -0.101 , 0.015). The raw unstandardized path coefficients are displayed in Fig. 3.³

Study 1 Discussion

While we found that participants in the autonomy-restricting ADM condition engaged in higher levels of unethical behavior, our results did not support that personal responsibility mediated this relationship. One reason for this could be a lack of specificity of the personal responsibility measure. That is, the measure of personal responsibility used in the current study referred to responsibility for the work participants did on the job, any consequences of the decisions of their job, whether customers' needs are met, and whether revenue was brought in for the hotel. It did not refer, specifically, to the responsibility for any *negative* outcomes associated with unethical behavior, such as disadvantaging the customer. Thus, we were concerned that this measure may be masking whether and how participants experience responsibility for a very specific negative outcome. Indeed, unethical behavior negatively affects others, and people are less likely to want to feel responsible for outcomes that they know adversely affect others (Jones, 1991). Thus, to gain a clearer picture of whether and how personal responsibility might explain the relationship between autonomy and unethical behavior in Study 2, we incorporated a more focused measure of responsibility, which aimed to specifically gauge how participants perceived themselves, versus

authority figures, as responsible for customers' feelings of being disadvantaged by participants' email responses. The more specific measure of responsibility also helps us examine whether autonomy-restricting ADM facilitates the moral disengagement mechanism of displacement of responsibility.

We further highlight a possible limitation with the wording of the autonomy-restrictive manipulation. Specifically, being told that it was in participants' best interests to use the Top Reply, may have facilitated feelings of coercion or a sense that failing to follow bureaucratic processes may lead to the individual being punished in some way for lack of compliance (Barker, 1993). To safeguard against signaling coercion, there were no signals included about punitive actions should participants deviate from the ADM-recommended answers. Furthermore, we do not suspect coercion played a strong role because many participants selected the more ethical alternative to the Top reply than one would expect had the manipulation been highly coercive (see Table 2 that the percentage of participants in the low-autonomy condition choosing a Top reply for unethical dilemma emails ranges from 48% to 70%). Nevertheless, in Study 2, we revised our manipulation of autonomy-restriction to further reduce risks of inducing a feeling of coercion by softening the tone of the low-autonomy manipulation to include focused language from Morgeson and Humphrey's (2006) measure of job autonomy.

Study 2

This study was pre-registered on aspredicted.org, available here: <https://aspredicted.org/blind.php?x=pt7k4s>, with deviations detailed in supplementary materials. The online supplement also contains information about our validation of the modified autonomy manipulation.

Method

The methodology of Study 2 was identical to that of Study 1 (see Fig. 2) in terms of sequence, with three minor changes to the content. Firstly, in the preamble of Riley Watson's email to CSOs, Riley frames the investment in SmartAnswer™ as a useful tool that aims to help CSOs do their job more efficiently and help them meet their performance indicators (see Table 4). The email also requests CSOs to stay alert for SmartAnswer™ recommended top replies that seemed nonsensical or unrelated to the query of the customer. This new instruction aimed to enhance the task's realism and provides participants (particularly in the autonomy-restrictive condition) a reason to carefully examine the "top reply" content. In two distractor emails, the original "top reply" response from Study 1 was replaced by a nonsensical response option that did not directly address the customer's

³ Conventional fit indices cannot be computed in Mplus when a path model includes count variables as dependent variables. Therefore, we ran the same model again while treating unethical behavior as a continuous dependent variable; the results suggested the model was likely well-fitting: $\chi^2(1, N = 276) = 3.48$, $p = .06$.

Table 4 Specific wording of autonomy condition manipulation for Study 2

Condition	Wording
Autonomy-affording ADM condition	To help you efficiently <i>respond to customer emails</i>, SmartAnswer™ will recommend a “top reply” to use (denoted by: ). <i>You can choose this top reply, or you can use your personal initiative and judgment to choose an alternative response. We encourage you to make your own decisions about how you respond to customers</i> You might have to deviate from the top reply if SmartAnswer™ presents responses <i>that are unrelated</i> to queries posed by a customer. Please stay alert for such instances. The SmartAnswer™ software is still in a trial stage and you can help maximize its efficiency by picking up these errors in these early stages
Autonomy-restrictive ADM condition	To help you efficiently <i>respond to customer emails</i>, SmartAnswer™ will recommend a “top reply” to use (denoted by: ). <i>You should choose this top reply, rather than using your personal initiative and judgment when responding to emails. We advise against making your own decisions about how you respond to customers</i> You should deviate from the top reply if SmartAnswer™ presents responses <i>that are unrelated</i> to queries posed by a customer. Please stay alert for such instances. The SmartAnswer™ software is still in a trial stage and you can help maximize its efficiency by picking up these errors in these early stages

query, while the alternative was sensible and aimed to positively meet KPIs. Secondly, we amended the wording of both autonomy manipulations to be more consistent with the language used to define and measure job autonomy in the Work Design Questionnaire (Morgeson & Humphrey, 2006). A final content change from Study 1 was a change made to the responsibility measure, explained below.

We recruited 268 individuals (49% female) from Amazon’s MTurk platform via CloudResearch. All participants were residents of the United States of America. Using our pre-registered data screening and exclusion criteria, we excluded 12 participants, leaving a final sample of 256. In the final sample, participants’ mean age was 38.74 years ($SD = 12.20$) and 82% of participants were employed in

either full-time, part-time, or casual work. Participants were paid US\$5 for their time.

All measures were the same as per Study 1 except that instead of using the personal responsibility measure from Study 1, we created a measure of participants’ “attribution of responsibility.”

Attribution of Responsibility

Participants were asked to imagine that they had been working as a CSO for a while, and that they recently learned that some customers contacted the hotel to complain about the responses to emails, which had made these customers feel unfairly disadvantaged. Participants rated the extent to which they attributed responsibility for how those customers were feeling to four agents. These agents were: the participant themselves (i.e., the CSO), the managers/executives at Paradise Point Hotel (i.e., the human figures of authority), the SmartAnswer™ software, and the designers of the SmartAnswer™ software. All entities were rated using a five-point Likert scale ranging from 1 (*should not be held responsible at all*) to 5 (*should be held completely responsible*). This method of rating responsibility is in line with the previous research that aims to measure task-specific levels of responsibility (Erlandsson et al., 2015) or blame (Ames & Fisk, 2013; Sullivan & Fosso Wamba, 2022). Personal responsibility was operationalized as the responsibility rating in relation to the self. Attribution of responsibility to authority was operationalized as whichever rating among the three figures of authority was highest. Thus, in effect, this measure is indifferent to *who* the responsible agent was and reflects the level of responsibility assigned to any agent, *other than the self*.

Results

An independent-samples *t*-test on the manipulation check item indicated a successful manipulation. Participants in the autonomy-restrictive condition ($N = 132$, $mean = 2.87$, $SD = 1.26$) experienced significantly lower levels of autonomy than to those in the autonomy-affording condition ($N = 123$, $mean = 4.06$, $SD = 0.78$), ($M_{diff} = 1.19$, $t(221) = -9.11$, $p < 0.001$, Cohen’s $d = 1.13$).⁴

As previously, unethical behavior was operationalized as a count variable, as per Study 1. Table 2 provides the percentage of participants that chose the “top reply” option in each condition; the response patterns in this Study are similar to those of Study 1. Table 2 shows that participants in the autonomy-restrictive condition were roughly two to

⁴ Levene’s test of equality of variances was violated ($F = 83.79$, $p < .001$), hence we report the *t*-test that accounts for the violation.

eight times more likely than those in the autonomy-affording condition to choose the SmartAnswer™ “top reply” option for the dilemma email stimuli. However, for most of the distractor stimuli, participants in the autonomy-restrictive condition were almost equally likely to choose the “top reply” as those in the autonomy-affording condition. A very low proportion of participants in either condition chose the nonsensical top replies corresponding to two of the distractor dilemmas.

Table 5 shows the descriptive statistics and correlations for Study 2. The table shows a negative relationship between assigned job autonomy condition and unethical behavior count, and personal responsibility_{T1} was negatively correlated with unethical behavior. Autonomy condition and personal responsibility were negatively correlated with attributing responsibility to authority, suggesting that autonomy-restrictive ADM was associated with higher levels of attribution of responsibility to an authority figure. Neither honesty-humility nor conscientiousness were associated with both the mediator (personal responsibility_{T1}) and dependent variables (unethical behavior, personal responsibility_{T2}, and attribution of responsibility_{T2}), therefore, neither variable was controlled for in the hypotheses tests.

To test Hypotheses 1 and 2, we specified the path model shown in Fig. 4, using Mplus version 8.3, which incorporates (a) the regression of unethical behavior on job autonomy condition and personal responsibility_{T1} and (b) the indirect path of autonomy condition on unethical behavior, via personal responsibility_{T1}. The indirect path was estimated by obtaining 95% bias-corrected confidence intervals derived from 10,000 bootstrapped samples (Hayes, 2018). We again discovered that the assumption of equidispersion of unethical behavior was violated ($\mu = 1.21$ being lower than $\sigma^2 = 2.12$) and thus estimated the unethical behavior regression paths using a negative binomial distribution.

Hypotheses 1 and 2 were supported. The results revealed a direct negative relationship between assigned autonomy condition on unethical behavior ($B = -1.10$, $SE = 0.17$, $p < 0.001$), and evidence of an indirect effect from autonomy condition to unethical behavior through personal responsibility_{T1}. The confidence intervals for the indirect pathway from autonomy to unethical behavior did not cross zero (indirect effect estimate = -0.083 , 95% CI = -0.172 , -0.029). In other words, participants who experienced autonomy-restriction (autonomy-affordance) subsequently experienced low (high) levels of personal responsibility_{T1} for the negative outcomes of their work, and in turn engaged in high (low) levels of unethical behavior. The raw unstandardized path coefficients from this analysis are displayed in Fig. 4. In contrast to Study 1 (in which personal responsibility did not mediate the relationship between autonomy condition and unethical behavior) in Study 2, personal

Table 5 Descriptive statistics and correlations among Study 2 variables

Variable	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. Autonomy condition ^a	NA	NA														
2. Unethical behavior count	1.21	1.46	-0.44													
3. Personal responsibility _{T1}	3.07	1.13	0.25	-0.24												
4. Personal responsibility _{T2}	3.07	1.14	0.29	-0.32	0.71											
5. Manager responsibility _{T1}	3.29	1.12	-0.22	0.12	-0.17	-0.11										
6. Manager responsibility _{T2}	3.37	1.12	-0.29	0.19	-0.16	-0.18	0.76									
7. Software responsibility _{T1}	2.91	1.16	-0.28	0.16	-0.03	-0.08	0.32	0.32								
8. Software responsibility _{T2}	3.06	1.18	-0.27	0.21	-0.10	-0.12	0.30	0.35	.74							
9. Software designer responsibility _{T1}	3.06	1.13	-0.28	0.20	-0.16	-0.12	0.44	0.41	0.64	0.61						
10. Software designer responsibility _{T2}	3.05	1.12	-0.31	0.22	-0.22	-0.22	0.45	0.54	0.57	0.70	0.77					
11. Attributing responsibility to authority _{T1}	3.70	1.02	-0.30	0.16	-0.18	-0.14	0.79	0.67	0.61	0.51	0.68	0.61	(0.73)			
12. Attributing responsibility to authority _{T2}	3.71	1.03	-0.35	0.24	-0.17	-0.21	0.65	0.80	0.52	0.62	0.57	0.68	0.80	(0.77)		
13. Honesty-humility	3.46	0.69	-0.13	0.03	-0.09	-0.05	-0.03	-0.05	0.05	0.10	0.02	0.08	0.01	0.06	(0.77)	
14. Conscientiousness	3.82	0.60	-0.07	0.05	-0.03	0.05	0.03	0.00	-0.08	-0.03	0.04	0.01	-0.01	-0.01	0.33	(0.81)

N = 256. Significant correlations ($p < .05$) are indicated in bold. Cronbach's coefficient alpha, where applicable, is provided along the diagonal in parentheses. All responsibility measures except for Attributing Responsibility to Authority are single-item measures. ^aAutonomy-restrictive ADM = 0, autonomy-affording ADM = 1

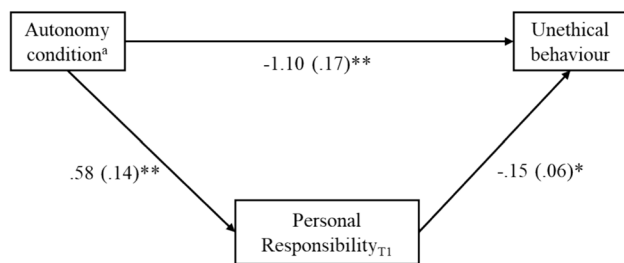


Fig. 4 Path model evaluating the indirect effect of autonomy condition on unethical behavior through personal responsibility_{T1}. Path coefficients are unstandardized, accompanied by standard errors in parentheses. *denotes $p < .05$; **denotes $p < .001$. ^aAutonomy-restrictive ADM=0, autonomy-affording ADM=1

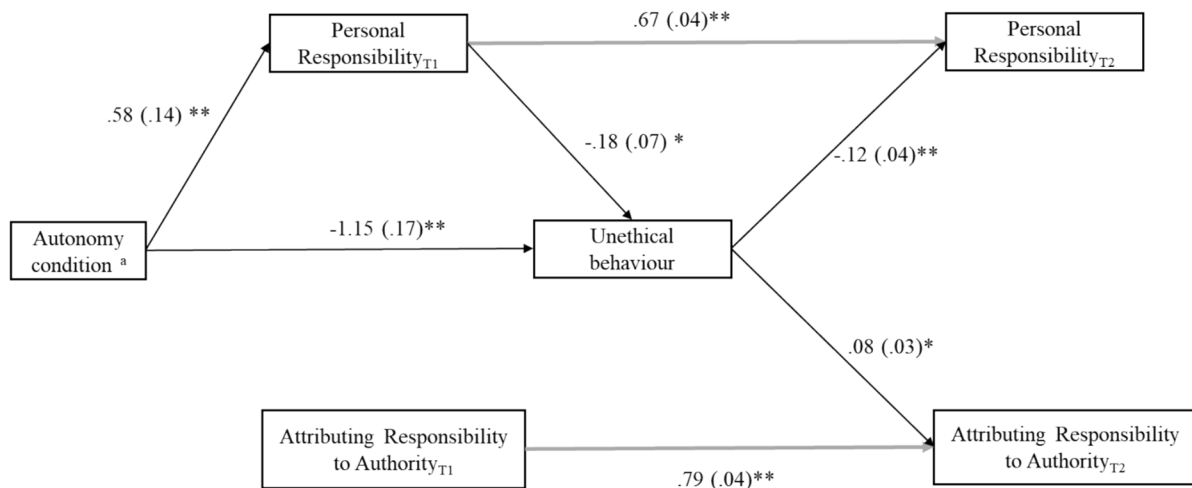


Fig. 5 Path model and unstandardized coefficients evaluating the causal effect of autonomy condition on displacement of responsibility onto a figure of authority. Note. *denotes $p < .05$; **denotes $p < .01$.

responsibility for the negative outcomes of one's work partially mediated this relationship.

To test Hypothesis 3, which focuses on what happens after unethical behavior has occurred, we specified the following path model (see Fig. 5) in Mplus version 8.3: (a) the regressions of personal responsibility_{T1} and unethical behavior on autonomy condition; (b) the regression of unethical behavior on personal responsibility_{T1} and Autonomy condition; and (c) the regressions of personal responsibility_{T2} and attributing responsibility to authority_{T2} on unethical behavior. We also controlled for attributing responsibility to authority_{T1} in the regression of attributing responsibility to authority_{T2} to enable capturing any changes in the extent to which participants attributed responsibility for negative outcomes to others, after having engaged in unethical behavior. Similarly, we also controlled for personal responsibility_{T1} in the regression of personal responsibility_{T2} to capture

changes in the level of personal responsibility attributable to unethical behavior.

In this analysis, unethical behavior was an outcome *and* predictor variable, hence, we treated the unethical behavior variable as a continuous variable in the path model. However, to ensure that the test of the path estimates in the final model were robust, we also conducted a separate analysis where we specified a negative binomial regression of unethical behavior on responsibility and autonomy. The two estimates were very similar to those obtained in the final model, suggesting that treating unethical behavior as a continuous variable for the purposes of these final analyses was not leading to misleading conclusions.⁵

The final model was an excellent fit for the data, χ^2 (6,

All other paths non-significant. Gray lines and font indicate relationships or variables included for control purposes. ^aAutonomy-restrictive ADM=0, autonomy-affording ADM=1

$N = 256$) = 8.79, $p = 0.186$; CFI = 0.99; RMSEA = 0.04; SRMR = 0.04, and the path estimates, shown in Fig. 5, also supported Hypothesis 3. Engaging in unethical behavior when working with autonomy-restrictive ADM was negatively associated with subsequent perceptions of personal responsibility_{T2} ($B = -0.12$, $SE = 0.04$, $p < 0.001$) and positively associated with subsequent attributing responsibility to authority_{T2} ($B = 0.08$, $SE = 0.03$, $p = 0.002$). These findings indicate autonomy-restriction led individuals to *displace* responsibility from themselves onto a figure of authority for the negative outcomes of their work, consistent with Hypothesis 3.

⁵ The unstandardized estimate for the regression of unethical behavior onto personal responsibility_{T1} when treating unethical behavior as a count variable was .03 lower here than in the final model, but the estimate was still significant ($p = .02$).

General Discussion

In this paper, we explored how individuals perceive responsibility for the ethical consequences of working with technology implemented in a way that either affords or restricts human job autonomy. Across our studies, we found that individuals working with autonomy-restricting ADM were more likely to engage in unethical behavior than those working with autonomy-affording ADM. Our research represented one of the first attempts we are aware of to capture the cognitions associated with displacement of responsibility and thus how this mechanism of moral disengagement might unfold in the context of working with ADM. Our findings are consistent with Bandura's (1986) propositions that suggest individuals are more likely to displace responsibility to figures of authority that are perceived to have the most legitimate control over the individual's actions. Further, we showed unethical behavior to be an outcome of the autonomy–responsibility relation in the Job Characteristics Model. Specifically, in Study 2, we found that autonomy-restrictive ADM predicted unethical behavior via low levels of personal responsibility for negative outcomes of one's work. The results of Study 2 further provided empirical support for the causal role of autonomy-restrictive ADM in triggering cognitions associated with the moral disengagement displacement of responsibility mechanism. We encourage future researchers to continue to examine unethical behavior and cognition through the lens of work design, especially as work redesign could be an effective intervention for unethical behavior.

Theoretical and Practical Implications

Implications for the Moral Consequences of Working with ADM

Our studies add to existing schools of thought that underscore the importance of carefully considering how new technology is designed and implemented into the workplace by organizations (e.g., Grote et al., 2024; Kellogg et al., 2020; Martin, 2019; Parker & Grote, 2022). We add to emerging empirical evidence that aims to understand how individuals experience responsibility when working with (Sullivan & Fosso Wamba, 2022) or designing ADM systems (Nichol et al., 2024). Our finding that employees will experience lower levels of personal responsibility for negative impacts on customers when working with autonomy-restrictive ADM suggests that these technologies should be designed to provide employees with appropriate levels of control over the ADM (so that they can override unethical recommendations). If employees are required to work with autonomy-restrictive ADM, then the party considered responsible for

any potential harms should be made explicit to the employees. Further, Grote et al., (2024) argued that if the technology has been designed to limit human input or autonomy, then the designer of the ADM should be held accountable for negative consequences arising from its use. Managers or executives who choose to implement autonomy-restrictive ADM technologies that aim to prioritize organizational efficiency at the expense of customers' best interests (like in the Australian Robodebt scandal; Holmes, 2023) perhaps need to be willing to hold *themselves* responsible for outcomes that are beyond their employees' control. Indeed, ultimately, technology cannot itself comprehend or feel responsibility towards humans. Additionally, it would be critical to ensure that the technology design does not prioritize organizational efficiency over the interests of customers.

Implications for Understanding Moral Disengagement

We further contribute to a critical gap in the research on the measurement of specific moral disengagement mechanisms (Schaefer & Bouwmeester, 2021). Indeed, our studies were the first that we know of to have systematically examined how working with ADM systems can facilitate moral disengagement, also having 'unpacked' and examined the individual theoretical cognitive components of displacement of responsibility. We add to a growing body of literature showing that specific contexts can facilitate specific mechanisms of moral disengagement (Barsky, 2011; Huang et al., 2018; Kish-Gephart et al., 2014). By focusing on job autonomy as an antecedent of moral disengagement and unethical pro-organizational behavior through the lens of Hackman and Oldham's (1976) classic Job Characteristics Model, this research has important implications for further delving into work design theory to examine moral disengagement in organizations. Indeed, research has not considered work design to function as a core facilitator of moral disengagement, despite ample evidence that the way in which employees experience autonomy at work can uniquely affect their sense of responsibility at work (e.g., Fuller et al., 2006; Humphrey et al., 2007).

Applied organizational research on work redesign interventions has also shown job autonomy to be a key vehicle for improving employee outcomes such as job performance, psychological contract fulfillment, and employee wellbeing (Holman & Axtell, 2016; Holman et al., 2010). Drawing upon work design theory may therefore help organizations understand how they can prevent or intervene on employees morally disengaging from unethical behavior. For example, future research could test how and why improving job autonomy (from existing low levels of job autonomy) as an organizational intervention or implementing autonomy-affording technologies from the outset can facilitate ethical decision making.

Limitations and Further Research

While we went beyond existing research that uses adapted trait-based measures to examine displacement of responsibility (by using an externally valid experimental design), our research design still focuses on hypothetical opportunities to engage in unethical behavior in a hypothetical implementation of ADM. We recognize that the laboratory approach raises questions about the extent that participants' sense of responsibility would be activated to the same extent that it would be in organizational settings. That is, participants' perceptions of responsibility for the hypothetical adverse consequences for customers in the simulation may not be equivalent to, and perhaps weaker compared to the analogous perceptions of responsibility employees might feel if in a similar job in a real organization. Although the experiments were designed to simulate working with ADM as closely as possible to the 'real world' and provide participants with targeted information about customers' hypothetical reactions to the CSO's email replies, future research should examine the relationships investigated here within organizations, using real technologies, and across job sectors. We must acknowledge, however, that such an approach may introduce practical and ethical challenges such as having people report on unethical behavior, which is known to be susceptible to socially desirable responding (Fernandes & Randall, 1992) and therefore care should be taken to encourage honest responses (e.g., emphasizing participant and organization confidentiality or anonymity). Another limitation regarding generalizability is that our studies were conducted with a western demographic sample, and therefore the relationships that emerged cannot be assumed to be universal. Prior research shows that the effects of work characteristics (including job autonomy) on work outcomes can be moderated by culture; that is, employees in non-western countries may not perceive high job autonomy to be as beneficial for their work as those in western countries (Erez, 2010). Therefore, future research should seek to examine the relationships investigated here in cross-cultural contexts, particularly as ADM technology is being implemented into workplaces globally.

Although our study used a novel method to unpack the *cognitions* associated with displacement of responsibility, moral disengagement also occurs in part to avoid feeling negative *emotions* associated with unethical behavior such as guilt or shame (Bandura, 1986). However, we did not measure the affective component of displacement of responsibility, hence, we suggest that future researchers investigate the interplay of the affective and cognitive components of displacement of responsibility. Our method of measuring displacement of responsibility could be adapted in future research to better understand the affective components of displacement of responsibility, by including measurements

of moral emotions like shame and guilt along with personal responsibility. That is, moral disengagement alleviates individuals from feeling unpleasant emotions associated with unethical behavior like shame and guilt. We would expect that along with the cognitive aspect of shifting personal responsibility from the self to a figure of authority, individuals who experience low job autonomy would also feel low levels of guilt or shame after engaging in unethical behavior.

Additionally, the way that we measured displacement of responsibility in Study 2 cannot be assumed to *completely* capture the components of this moral disengagement mechanism. That is, we could not isolate exactly *when* participants felt that responsibility for the negative outcomes of their work should shift away from themselves and onto a figure of authority. In an inbox task like that employed in this study, the closest one might get to measure the moment this shift occurs would likely require asking participants about their perceptions of responsibility after each email they work through. Doing so, however, would risk creating demand characteristics for participants and may bias their responses during the task. Further, in a review of moral disengagement and its measurement, Moore (2015) acknowledged that for researchers to capture moral disengagement "in the moment" (p. 202) brain imaging technologies like functional magnetic resonance imaging may be required to provide researchers with a more complete picture of the process of moral disengagement.

The persistent direct effect between autonomy-restrictive ADM and unethical behavior across both studies suggests that future research should investigate other explanatory mechanisms in the relationship between autonomy and unethical behavior; one such mechanism may be self-efficacy. Self-efficacy refers to an individual's belief that they can carry out actions effectively (Bandura, 1986) and studies have shown that job autonomy can positively influence employees' sense of self-efficacy (e.g., Parker, 1998). Parker (2014) and Hannah et al. (2011) suggested may influence individuals' motivation to engage in moral behaviors. In the context of ADM, the technology relieves humans of the opportunity to reason through decisions themselves (Kellogg et al., 2020). Thus, using ADM technology could make employees feel that they are not as capable of making moral decisions or deviating from ostensibly unethical directives from figures of authority.

Another explanatory mechanism could be employees' role orientation, which refers to how employees "define their work role, including what types of and breadth of tasks, goals and problems they see as within their set of responsibilities, and how they believe they should approach those tasks, goals, and problems to be effective" (Parker, 2007, p. 404). Prior research shows lower job autonomy is associated with narrower role orientations and thinking that certain tasks and responsibilities are beyond the scope of one's

job (Parker et al., 1997). Autonomy-restricting ADM could facilitate unethical behavior if employees perceive that their job involves only doing what they are told to perform effectively (i.e., adhere to recommendations of ADM), or that dealing with the (short and long term) outcomes of unethical behavior are beyond their job scope.

Finally, we encourage researchers to examine moderators to understand what could buffer against unethical behavior when humans work with autonomy-restrictive ADM, such as transparency of the algorithm or humans' moral awareness that they may be actioning an unethical recommendation (e.g., Ebhrahimi & Matt, 2023). If an algorithm is designed in a way that has to provide humans with their reasoning of why a particular course of action is suggested, and the reasoning is not entirely justifiable, employees with lower levels of job autonomy may be more inclined to exercise their choice and deviate from the algorithm's recommended action.

Conclusion

The use of algorithmic decision making in organization can have significant ethical consequences for the organization's clients, as demonstrated by the Robodebt scandal (Holmes, 2023); these technologies need to be implemented in a way that supports, not hinders ethical decision making. The studies presented here highlight the importance of implementing autonomy-affording ADM for employees to reduce the likelihood of unethical behavior and displacement responsibility when working with algorithmic technology. Altogether, we hope our research encourages further lines of inquiry into leveraging good work design to facilitate ethical behavior and cognitions, as organizations increasingly implement technology into the workplace.

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Declarations

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Informed Consent All participants involved in this research provided informed consent to participate.

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